Cal-Adapt: ANALYTICS ENGINE

Co-producing state-level climate services — an operational framework from the Cal-Adapt: Analytics Engine

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Need for co-produced climate services

- *The use of climate information in planning and decision making remains low*, despite the proliferation of climate data, tools, and decision-support platforms[1,2].
- *Climate services often do not reflect how practitioners make decisions.* They are frequently produced without awareness of the existing workflows and practical constraints that practitioners face[3].
- *There is a need to collaboratively develop climate services alongside data users.* However, examples of how to integrate climate science and operational expertise into usable climate services remain rare[4,5].

State-level framework for the United States

- We present a state-level framework for co-producing climate services in the United States. It draws from four years of qualitative research on the Cal-Adapt: Analytics Engine (AE) and key findings from climate services and co-production literature.
- We aim to provide a practical starting point for other states and regions that want to co-produce transparent and decision-relevant climate services.

The framework identifies three components that are critical for co-producing climate services:

1. Assembling an interdisciplinary project team
2. Structuring an iterative co-production process
3. Designing products to meet diverse user needs and evolving workflows

Case study – Cal-Adapt: Analytics Engine

- The Cal-Adapt: Analytics Engine (AE) is a state-level, co-produced climate services platform developed for California's electricity sector.
- The AE contains high-resolution downscaled datasets, advanced analytical tools, customizable Jupyter notebooks (Fig. 1), and detailed guidance materials to support electricity sector planning and adaptation.
- The AE was developed through extensive co-production between climate scientists, social scientists, developers, consultants, and practitioners from electric utility companies (IOUs, POUs), state agencies, and local governments (Fig. 3).

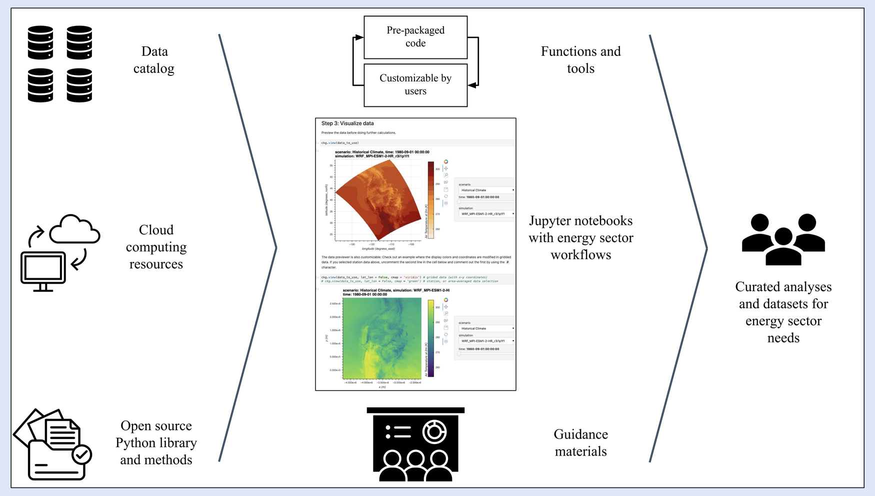

Figure 1. Components of the AE that were collaboratively developed to facilitate access and use of customizable and decision-relevant information.

Assembling an interdisciplinary project team

- Developing usable climate services requires assembling interdisciplinary project teams[1,6]. The AE relied on an interdisciplinary project team structure that included:
  - **Functions** – distinct, but interconnected project objectives (Fig. 2)
  - **Roles** – individuals' primary topical expertise and responsibilities. Over time, most folks also took on secondary or tertiary roles that required them to develop interdisciplinary expertise (Fig. 2).
  - **Sub-teams** – individuals from various roles were brought together to fulfill different project functions (not included in Fig. 2).

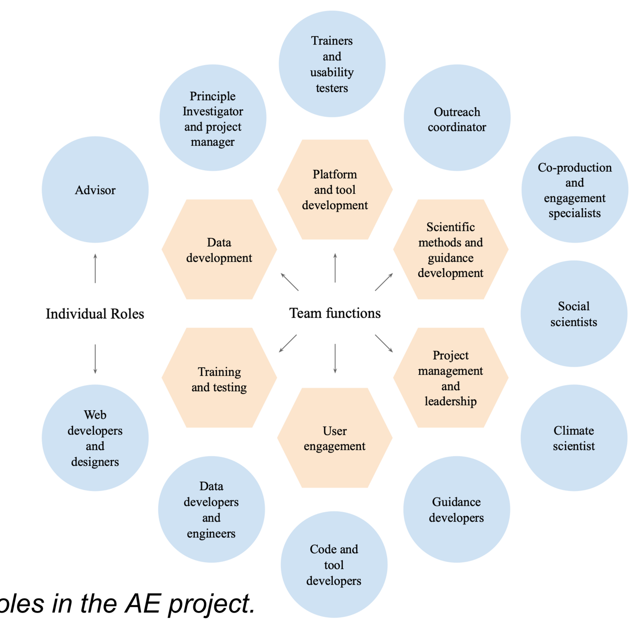

Figure 2. Functions and roles in the AE project.

Structuring an iterative co-production process

- An equally important aspect of co-producing climate services is determining how to conduct transdisciplinary work between an interdisciplinary project team and various partners[7,8]. The AE engagement strategy relied on a three-part structure:
  - **Tiered user groups** – three user groups that drove AE development priorities at different times. User groups were categorized by level of technical expertise, influence over decision processes, and available time for the project (Fig. 3).
  - **Use-case development stages** – six stages that were operationally used to transform a given climate data need (a use-case) into a climate services tool, workflow, or other type of output (Table 1).
  - **Engagement activities** – external transdisciplinary engagement activities brought specific user groups into different stages of use-case development; internal interdisciplinary engagements were conducted within the project team (Fig. 4).

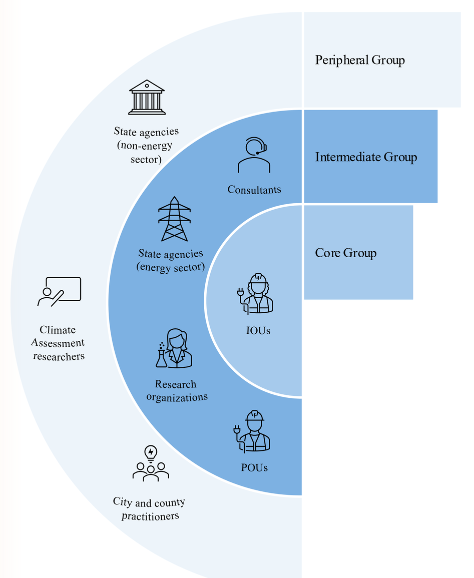

Figure 3. Tiered AE user groups.

Use-case development stage	Associated steps
Formulate the problem	Identify the problem Assess current and future needs
Bound the problem	Scope the problem space Identify integrated possibilities and constraints Identify a common output or outcome Agree upon design strategy
Design solutions	Develop methods Develop prototypes QA/QC process
Review solutions	Present work in progress Conduct usability testing Broaden functionality of methods and prototypes Refine methods and prototypes
Finalize solutions	QA/QC final products Deploy final products
Monitor and evaluate	Evaluate process and limitations Plan for updates and improvements Re-assess current and future needs

Table 1. Six stages of use-case development that can be implemented linearly, iterated between, and/or modified for the problem context.

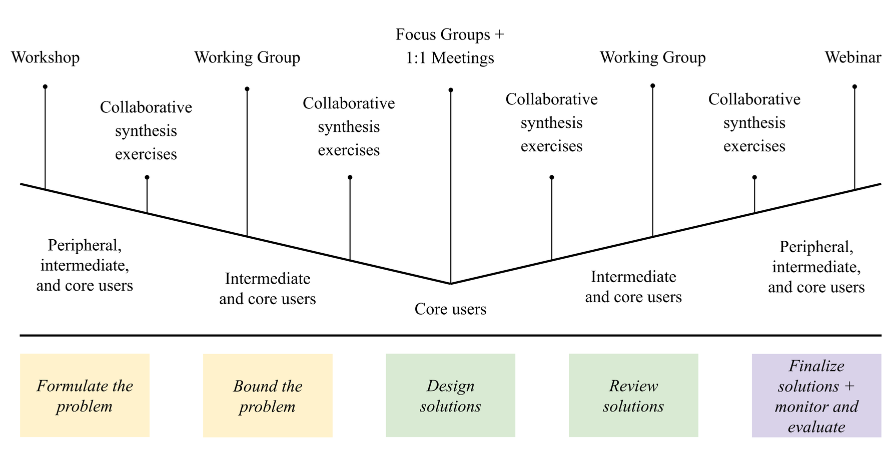

Figure 4. Engagement activities. Use-case development stages (bottom) and user groups (middle) are combined through external and internal engagement activities (top).

Designing products to meet diverse needs and evolving workflows

- Determining which products to build and what functionalities users need are critical aspects of creating usable climate services. State-level products must account for different operational contexts and varied levels of expertise of working with climate data[9,10].
- AE climate services were designed to be flexible and scaffolded to accommodate different technical capacities, workflows, and planning contexts. This included development of:
  - **Basic, intermediate, and advanced products** – designed for all tools, workflows, guidance, and notebooks.
  - **Exploratory and explanatory products** – designed to help users compare analyses, build intuition, and answer specific planning questions.
  - **Guidance materials** – designed to help users move from identifying a question, to developing an analysis, and then devising a solution.

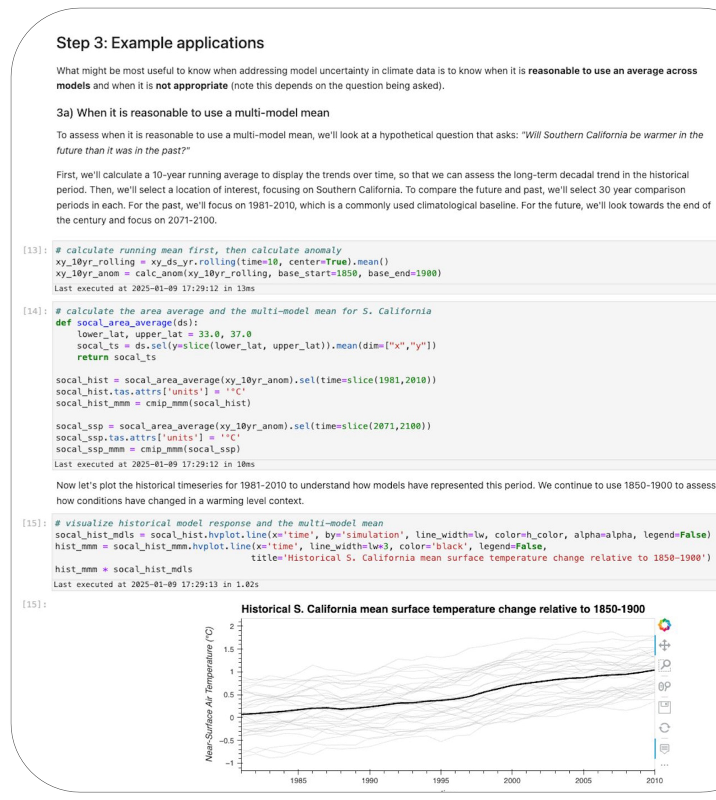

Figure 5. One section of an AE Jupyter notebook, illustrating guiding information, editable code, and associated visualizations of climate data.

Challenges and limitations

- *Balancing engagements with different user groups.* Regular discussion and negotiation are required to appropriately meet different user needs.
- *Translating user feedback into actionable development steps.* This often requires multiple rounds of user feedback synthesis, identification of needs, deliberation, and prioritization of next steps.
- *Sustained funding and a stable boundary organization.* These are core components of developing a multi-year co-production project that can meet evolving science and user needs.

Implications for co-produced climate services in the United States

- Co-produced climate services are critically needed as climate risks increase and planning pressures intensify.
- *Our framework provides a practical blueprint to help states build climate services that are decision-relevant and flexible for different users.* It should be adapted to regional institutional, political, and funding contexts.
- As states undertake similar efforts, we call for a national consortium of climate services organizations that can support cross-regional learning and shared methodological advances.

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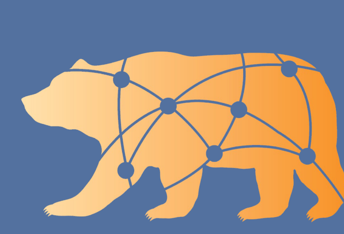

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